Numerical Analysis Quiz Name: _____

Question 1.

Let f be a sufficiently smooth function and let $h > 0$. Consider the following approximation to $f'(x)$:

$$D_h f(x) = \frac{af(x-h) + bf(x) + cf(x+2h)}{h}.$$

(a) Using Taylor expansions about x, determine the coefficients a, b, and c so that $D_h f(x)$ approximates $f'(x)$ with the highest possible order of accuracy.

(b) Determine the leading term of the truncation error $D_h f(x) - f'(x)$.

Question 2.

Let $f \in C^4([x-2h, x+2h])$ and let $h > 0$. Consider the following approximation to $f''(x)$:

$$D_h f(x) = \frac{f(x-2h) + 4f(x-h) - 10f(x) + 4f(x+h) + f(x+2h)}{8h^2}.$$

Using Taylor expansions about x with Lagrange remainders and the intermediate value theorem, show that there exists $c \in (x-2h, x+2h)$ such that

$$D_h f(x) - f''(x) = \frac{5}{24} h^2 f^{(4)}(c).$$

What is the order of accuracy of this approximation?